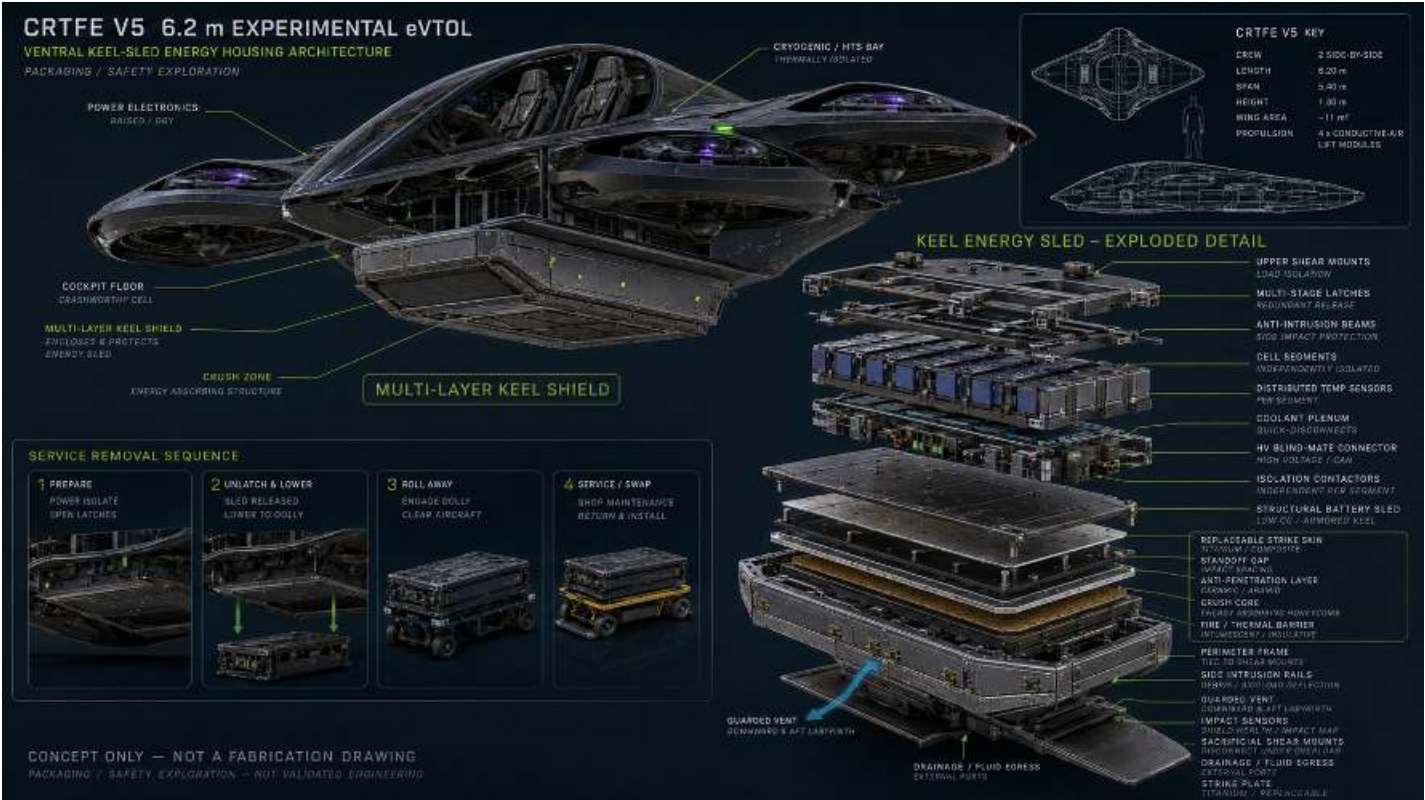


CRTFE V5

VENTRAL KEEL-SLED ENERGY HOUSING

Preliminary engineering definition, parts list, mockup build instructions, fastener schedule, calculations, and verification gates



CONCEPT ARCHITECTURE ONLY - NOT A FABRICATION DRAWING - NOT FLIGHT READY

Scope: This package defines a safe, non-energized full-scale packaging mockup and a preliminary design basis for later professional engineering. It does not authorize installation of live cells, high voltage, cryogenics, pressurized coolant, powered lift hardware, tethered flight, or crewed testing.

Revision	P0 - preliminary architecture	Date	2026-08-23
Vehicle basis	CRTFE V5 target concept	Design state	Unvalidated / research

1. Design basis and hard limits

The project data support a packaging study, not a flight design. The following values are carried directly from the supplied CRTFE V5 concept and V4/V5 simulation summary. Any value not listed here is an assumption or a preliminary candidate.

Parameter	Project value	Design consequence
Crew	2, side-by-side	Energy system must remain outside protected occupant volume.
Gross mass	650 kg	No mature growth reserve in current budget.
Vehicle envelope	6.20 m L x 5.40 m span x 1.80 m H	Keel sled must preserve ground clearance and egress.
Battery-system allowance	120 kg	Includes cells, enclosure, BMS, switching, cooling, shield and wiring unless budget is revised.
Central power bay	Approx. 1.60 m longitudinal station	Sled placed at/near CG; final station requires weight-and-balance model.
Working hover input	Approx. 228 kW total	Unvalidated placeholder; drives high-power bus and severe thermal demand.
Lift architecture	4 modules	Four inverter feeds and module-level fault isolation.
Cryogenic/thermal bay	Approx. 1.10 m, aft	Separate from battery bay and occupant cell.
Design status	Conditional simulation	No energized or flight build until test gates close.

Decision gate: The 120 kg allowance and 228 kW placeholder do not by themselves define a battery. Chemistry, voltage, usable energy, continuous and pulse C-rate, thermal rejection, reserve, and thermal-runaway behavior must be selected and tested before cell count or bus voltage can be released.

Preliminary packaging envelope

A removable sled envelope of **1.45 m x 1.05 m x 0.34 m maximum** is used only to coordinate space. Gross volume is 0.518 m3. It does not define usable cell volume; shielding, crush space, cooling, buswork, switching, clearances, mounts and vent passages consume a substantial fraction.

2. Math traceability and feasibility checks

Check	Equation / Input	Result	Interpretation
Weight	650 kg x 9.80665 m/s ²	6.37 kN	Matches supplied hover sanity check.
Module thrust	6.37 kN / 4	1.59 kN/module	Use updated 650 kg vehicle basis, not older 450 kg simulation baseline.
Energy - 5 min	228 kW x 5/60 h	19.0 kWh	Before reserve and conversion losses.
Energy - 10 min	228 kW x 10/60 h	38.0 kWh	Before reserve and conversion losses.
Energy - 15 min	228 kW x 15/60 h	57.0 kWh	Before reserve and conversion losses.
Illustrative 120 kg system	120 kg x 200 Wh/kg system x 80% usable	19.2 kWh usable	About 5.1 min at 228 kW; 200 Wh/kg is an assumption, not project data.
Power density	228 kW / 120 kg	1.90 kW/kg system	Cells, interconnects, cooling and contactors must all support this duty.
Sled gross volume	1.45 x 1.05 x 0.34	0.518 m ³	Packaging envelope only.

Mass-allocation working target (must be reconciled)

Subsystem	Working target	Notes
Cells/modules	75-85 kg	Selected only after chemistry and C-rate tests.
Structural sled + internal frames	10-14 kg	FEA and modal test required.
Multi-layer shield + crush core	10-16 kg	Likely mass driver; cannot be finalized without threat/load definition.
Cooling hardware and coolant	6-10 kg	Excludes vehicle heat exchanger if accounted elsewhere.
BMS, contactors, precharge, fusing, buswork	6-9 kg	Voltage and fault-current dependent.
Harness, sensors, seals, fasteners	3-5 kg	Includes service disconnects.
Total range	110-139 kg	Current 120 kg budget sits inside range but has little margin.

The mass range demonstrates why the concept package warned that the vehicle may move into a 750-900 kg class. Shield mass cannot be added without a full aircraft mass-budget update.

3. Blueprint-style architecture plate

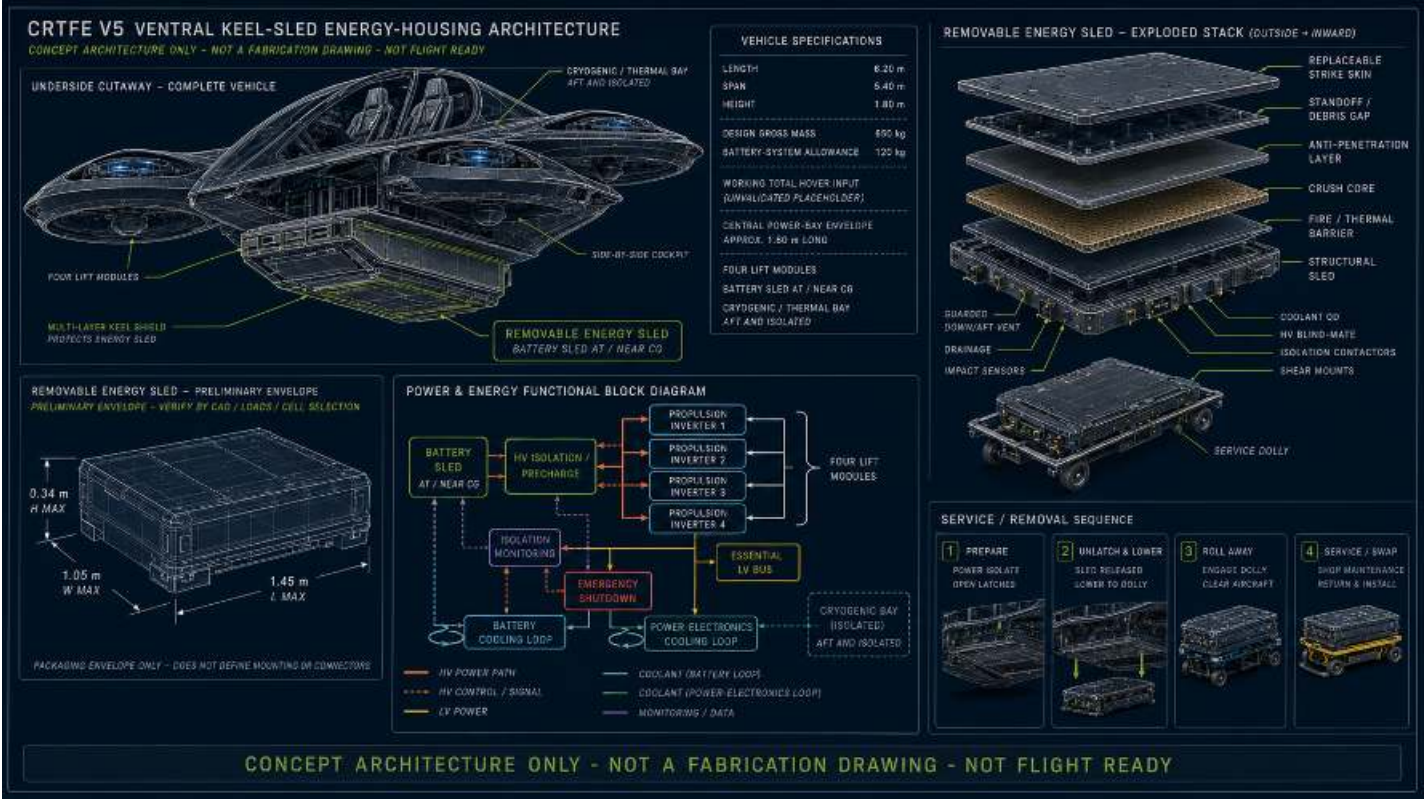


Figure 1. Packaging architecture derived from supplied project data. The 1.45 x 1.05 x 0.34 m sled is a preliminary coordination envelope only.

Interface zones

Zone	Required separation / function
Occupant cell	No cells, HV contactors, pressure relief, or vent discharge inside the crashworthy cell.
Battery sled	Segmented modules; service-disconnect state visible; independent monitoring and isolation.
Keel shield	Replaceable strike skin, standoff gap, anti-penetration layer, crush core, thermal barrier, structural sled.
Vent path	Guarded downward/aft route; must be proven by gas/particle/fire test and must not threaten occupants, controls, landing gear, or lift ducts.
Cooling	Battery and power-electronics loops segregated; leak detection and dry-break couplings at service boundary.
Cryogenic bay	Aft, thermally isolated; quench protection and energy dump remain separate development work.

4. Preliminary parts list - structure and shielding

	Qty	Material/Description/Requirement	Design/Performance/Disposition
S-001	1	Structural sled frame - 7075-T73 aluminum or Ti-6Al-4V local fittings	Welded aluminum is not assumed. Use machined/extruded members with isolated bolted joints; final alloy and temper by corrosion/FEA review.
S-002	1	Inner fire/thermal pan - Inconel 625 or 321 stainless thin sheet	Thickness TBD by fire and pressure test; electrically bonded/grounded as required.
S-003	1	Microporous insulation or aerospace ceramic-fiber blanket	Chemistry-compatible, low smoke/toxicity; validate compression and moisture behavior.
S-004	1	Aluminum honeycomb crush core	Nominal 25-50 mm study range; density/crush plateau selected by impact test.
S-005	1	Anti-penetration layer - aramid laminate with optional ceramic strike tiles	Threat definition and test coupons required; do not claim ballistic rating.
S-006	1	Standoff/debris frame - 7075-T73 rails	Creates inspectable gap; includes drain paths and vent labyrinth.
S-007	1	Replaceable strike skin - Ti-6Al-4V sheet or hybrid titanium/composite	Thickness TBD after impact spectrum, ground clearance and mass trades.
S-008	2	Side intrusion rails - 7075-T73 or Ti-6Al-4V	Terminate into sacrificial load paths, not battery modules.
S-009	8	Sacrificial shear-mount cartridges	Replaceable, serialized; release load and energy absorption TBD by test.
S-010	4	Multi-stage latches with positive visual lock indication	Ground mockup may use industrial draw latches; flight candidates require qualified aerospace hardware.
S-011	1	Perimeter environmental seal - fluorosilicone	Final compound must be electrolyte/coolant/fire-agent compatible.
S-012	1	EMI/bonding gasket	Continuity target and lightning/HIRF environment TBD.
S-013	8	Impact/strain sensor nodes	Tri-axial accelerometer or piezoelectric witness; measurement range selected after drop-test plan.
S-014	1	Service dolly, braked casters, mechanical lift table	Rated at least 2x mockup mass; never work beneath unsupported sled.
S-015	4	Hard-point lifting eyes / handling fixtures	Ground handling only until proof-loaded and formally approved.

5. Preliminary parts list - electrical, thermal, sensing

	Qty	Item	Comments
E-001	TBD	Battery modules/cells	No procurement release until voltage, energy, C-rate, thermal-runaway propagation and supplier quality plan are approved.
E-002	1	Battery management system	Cell voltage/temperature, current, insulation state, contactor weld detection, data logging and independent shutdown path.
E-003	2	Main HV contactors	Series isolation; voltage/current/fault rating TBD. Include economizer and auxiliary state contacts.
E-004	1	Precharge contactor + resistor assembly	Sized from actual bus capacitance and target precharge time; include discharge path.
E-005	1	Main fuse or pyrotechnic disconnect	Fault-current study and coordination required before selection.
E-006	1	Manual service disconnect	Tool-controlled, interlocked, touch-safe, visible state.
E-007	1	Isolation monitoring device	Continuous HV-to-chassis monitoring; thresholds set by bus voltage and safety assessment.
E-008	1 set	HV busbars and flexible links	Laminated/insulated; ampacity, creepage, clearance and short-circuit force analysis TBD.
E-009	1	Blind-mate HV connector	Finger-safe, first-mate/last-break interlock, environmental sealing; rating TBD.
E-010	1	Low-voltage essential bus interface	Galvanically isolated from propulsion bus; supports emergency shutdown and data recorder.
T-001	1	Battery cold plate / cooling manifold	Leak-tested; electrically isolated; flow distribution verified with instrumented dummy modules.
T-002	2	Dry-break coolant quick disconnects	Keyed supply/return, low spill, pressure rating TBD.
T-003	1	Pressure relief / expansion volume	Cooling-loop component; not the battery thermal-runaway vent.
T-004	1	Guarded thermal-runaway vent labyrinth	Cross-section and burst features TBD by gas-generation and propagation tests.
T-005	2	Leak sensors	Placed at low points and connector interface.
I-001	24+	Temperature sensors	At cells/modules, coolant inlet/outlet, shield cavity and vent path.
I-002	4+	Gas/smoke sensing points	Chemistry-specific detection study required.
I-003	1	Sled data recorder / event logger	Independent timestamped capture of BMS, isolation, impact and thermal data.
W-001	1 lot	LV harness, shielded twisted pairs, backshells and clamps	Separation from HV; service loop; chafe protection; bend radius and connector keying controlled.
W-002	1 lot	Bonding straps and conductive hardware	Bonding/grounding plan and corrosion isolation required.

6. Candidate fastener schedule

These selections are detailed enough to build a non-energized fit-check mockup. They are not released flight fasteners. Critical-joint diameter, grip, preload, torque and quantity require a joint load model, bearing/bypass analysis, pull-through testing, fatigue and vibration testing, and galvanic review. Torque changes with finish, lubricant, reuse and prevailing-torque nut condition.

F-01 Strike-skin removable panels	M6 x 1.0, A286 stainless, socket flange bolt	24	A286 large-area washer; A286 all-metal prevailing-torque nut	7-9 N.m dry	Panel flutter, impact, insert pull-out and vibration test.
F-02 Standoff frame to perimeter frame	M8 x 1.25, A286 stainless hex-flange bolt	16	A286 washer + all-metal locknut	20-24 N.m dry	Combined shear/tension analysis; calibrated torque-tension test.
F-03 Anti-intrusion side rails	M8 x 1.25, Ti-6Al-4V bolt where galvanically isolated	12	Ti or isolated A286 washer/nut; insulating sleeves at carbon interfaces	TBD by coupon	Bearing/bypass, galvanic, fatigue and impact validation.
F-04 Structural sled corner joints	M10 x 1.5, A286 or alloy-steel aerospace bolt candidate	16	Hardened washer + all-metal locknut	Do not torque for flight	Preloaded-joint analysis; proof and fatigue test.
F-05 Sled-to-airframe shear mounts	M12 x 1.75 equivalent preliminary envelope; A286/Inconel candidate	8	Spherical washer pair if misalignment requires; positive secondary retention	No provisional flight torque	Aircraft loads model, crash pulse, mount release curve, fail-safe assessment.
F-06 Internal module hold-downs	M6 x 1.0 A286 with insulated bushings	24-48	Large-area washer; locking nut; captive where possible	6-8 N.m dry	Cell/module supplier limits, dielectric isolation and shock test.
F-07 Cold plate to module rails	M5 x 0.8 A286	24+	Belleville stack only if analysis supports; all-metal locknut	3.5-4.5 N.m dry	Thermal interface pressure map and cycling test.
F-08 Contactor/BMS brackets	M5 x 0.8 A286 or 316 stainless	12-20	Flat washer + all-metal locknut	3.5-4.5 N.m dry	Vibration, electrical clearance and bracket fatigue.
F-09 Service panels	M5 x 0.8 A286 captive screw	20-32	Floating nutplate; conductive or isolated finish per bonding plan	3-4 N.m dry	Repeated-removal endurance and FOD retention.
F-10 Sensor brackets / P-clamps	M4 x 0.7 A286	20+	Washer + all-metal nut or keyed nutplate	1.8-2.2 N.m dry	Harness pull, chafe and vibration test.
F-11 Grounding/bond straps	M5 x 0.8 A286	8+	Toothed bonding washer only where approved; corrosion seal after test	Per bonding test	Milliohm target, lightning/HIRF and corrosion plan.
F-12 Dolly handling fixtures	M10 x 1.5 class 10.9 zinc-nickel steel, ground equipment only	16	Class 10 all-metal nut + hardened washer	45-50 N.m dry	Not aircraft hardware; proof-load dolly separately.

Fastener control notes: use lot traceability; record grip length and installed orientation; avoid threads in primary shear planes; use metallic/composite isolation sleeves where required; use calibrated tools; record running torque separately from final torque; do not substitute nylon-insert nuts near heat/fire zones; use witness marks only as inspection aids, not as the locking method.

7. Build instructions - non-energized packaging prototype

1	Freeze the mockup configuration Issue a drawing index, mass target, coordinate system, preliminary CG station and keep-out zones. Mark every part MOCKUP ONLY. Use inert dimensional cell blocks; no live cells, HV, cryogenics, pressurized coolant or pyrotechnic disconnects.
2	Build a full-scale interface fixture Construct a rigid fuselage/keel fixture representing the 1.60 m bay, cockpit floor, landing-gear keep-out, duct keep-out and service-dolly approach. Survey datum points before assembly.
3	Fabricate the structural sled Machine/assemble the 1.45 x 1.05 x 0.34 m maximum frame. Fit sleeves/inserts at composite interfaces. Deburr, edge-break, clean and inspect. Add temporary ballast pockets so mass and CG can be adjusted safely.
4	Install segmented inert modules Use dimensionally representative noncombustible blocks. Preserve proposed module gaps, cooling passages, sensor paths, HV clearance volumes and removal access. Record actual occupied volume.
5	Install dummy electrical and thermal interfaces Use nonconductive mock busbars, dead contactor envelopes, dead service disconnect, dummy coolant manifolds and keyed connector shells. Verify hand access, tool swing and interlock sequencing.
6	Assemble inner containment and shield stack Fit structural pan, thermal barrier, crush core, anti-penetration layer, standoff frame and strike skin. Use candidate F-01/F-02 fasteners. Do not adhesive-bond the prototype unless disassembly coupons are planned.
7	Integrate vent and drainage mockups Use smoke-safe visualization or low-pressure air only. Confirm the route exits down/aft and remains clear of cockpit, controls, landing gear, lift ducts and hot surfaces. Do not use combustion.
8	Fit shear mounts and latches Install instrumented or replaceable mock mount cartridges. Verify positive lock indication, secondary retention, no single-point release, and safe lowering onto the dolly.
9	Perform service trial With a qualified lift plan, isolate, unlatch, lower, roll away, inspect, reinstall and relock. Time each action. Confirm no person is under suspended hardware.
10	Run fit, stiffness and handling tests Use distributed proof loads and ballast only within a written test plan. Measure deflection, latch alignment, connector misalignment and dolly stability. Stop for cracks, yielding, binding or unexpected load transfer.
11	Update CAD and mass properties Weigh every subassembly, update the mass budget and envelope, and issue redlines. Do not proceed to energized hardware until the preliminary design review closes all open assumptions.
12	Advance through gated prototypes Coupon tests -> subscale enclosure -> instrumented inert full-scale sled -> low-energy cell module in a remote fire-test facility -> full containment test -> integrated iron-bird. Crewed installation is a separate later program.

8. Assembly manual - controls, tools and work area

Manual applicability

This procedure applies only to the full-scale inert packaging prototype. Replace all battery cells with dimensionally representative noncombustible blocks; replace HV conductors with nonconductive mock busbars; keep coolant lines dry or use low-pressure water during dedicated flow tests. A separate approved procedure is mandatory for any live cell, HV, pressurized, cryogenic, pyrotechnic, fire or propulsion work.

Control	Requirement
Minimum crew	Assembly lead plus one qualified technician. Use a third person for lifts, sled movement and independent inspection.
Document control	Use the current drawing/BOM revision. Redline deviations. Record serial/lot numbers for fasteners, inserts, latches, sensors and structural layers.
Foreign-object control	Inventory tools and loose hardware at start/end of every shift. Use covered parts bins and magnetic/nonmagnetic retrieval tools as appropriate.
Torque control	Use an in-calibration torque wrench covering the required range. Record tool ID, calibration due date, running torque, final torque and witness-mark status.
Lift safety	Use a written lift plan. Chock dolly wheels. Keep hands clear of pinch zones. Never place a person under unsupported hardware.
Material handling	Wear gloves for composites/aramid/ceramic. Control dust. Follow supplier SDS for sealants, coatings and cleaning materials.
Stop work	Stop for wrong part/revision, damaged threads, delamination, cracking, binding, misalignment, unexplained preload loss or any change to protected clearances.

Required tools and support equipment

Category	Minimum equipment
Measurement	Steel rules, digital calipers, angle gauge, feeler gauges, straightedge, laser level or survey instrument, calibrated scales.
Fastener installation	Metric hex/socket tools, open-end/ring spanners, calibrated 1-10 N.m, 5-30 N.m and 20-100 N.m torque wrenches, torque adapters, running-torque gauge.
Hole/insert work	Drill press or controlled drill guide, reamers, depth stops, countersink/deburr tools, insert installation tooling, vacuum extraction. Follow released mockup drawing.
Handling	Rated lift table/service dolly, slings/spreader where required, wheel chocks, soft jaws, nonmarring supports, temporary guide pins.
Inspection	Bright light, mirror/borescope, thread gauges, crack/damage magnification, continuity meter for bonding mockups only, camera and inspection tags.
Consumables	Lint-free wipes, approved cleaner, corrosion-isolation film/sleeves, fluorosilicone seal samples, removable thread witness lacquer, labels and serialized bags.
PPE	Eye protection, safety footwear, cut-resistant gloves, hearing protection for machining, respiratory/dust protection per material SDS.

9. Assembly manual - receiving and preassembly

A01	Verify documents Confirm BOM, fastener schedule, mockup drawings, revision status, inspection plan and lift plan. Write the revision on the traveler.	Builder: ____ Inspector: ____ Date: ____
A02	Prepare the work cell Clean the floor and benches. Establish a marked sled footprint, clean-parts table, quarantine area, FOD board and protected composite/ceramic cutting area.	Builder: ____ Inspector: ____ Date: ____
A03	Inventory parts Count and identify S-001 through S-015, E/T/I/W mock components and F-01 through F-12 hardware. Quarantine shortages and substitutions.	Builder: ____ Inspector: ____ Date: ____
A04	Inspect fasteners Reject damaged threads, mixed grades, missing material certificates, corrosion, burrs or unknown coatings. Verify size with thread gauges and grip stack with calipers.	Builder: ____ Inspector: ____ Date: ____
A05	Inspect structural members Check rails, corner fittings, plates and inserts for dents, cracks, warping, edge damage and blocked drain/vent features.	Builder: ____ Inspector: ____ Date: ____
A06	Inspect shield layers Check strike skin, standoff frame, anti-penetration panels, crush core and thermal barrier against templates. Do not install cracked ceramic or wet/damaged core.	Builder: ____ Inspector: ____ Date: ____
A07	Prepare datum fixture Level and secure the fuselage/keel interface fixture. Mark aircraft X/Y/Z axes, sled origin, CG reference, landing-gear keep-out and removal corridor.	Builder: ____ Inspector: ____ Date: ____
A08	Set up the dolly Chock wheels, verify brake function, center the lift table and install nonmarring supports. Confirm rated load is at least twice the inert mockup mass.	Builder: ____ Inspector: ____ Date: ____
A09	Dry-fit structural frame Assemble S-001 members with temporary guide pins. Measure diagonals, twist and envelope before installing final mockup fasteners.	Builder: ____ Inspector: ____ Date: ____
A10	Record baseline dimensions Record length, width, height, diagonal difference, rail parallelism and empty-frame mass. Obtain inspector acceptance before continuing.	Builder: ____ Inspector: ____ Date: ____

HOLD POINT HP-1: Do not proceed until the empty frame is dimensionally accepted, all parts are identified, and the dolly/lift setup is approved.

10. Assembly manual - structural sled and interfaces

A11	Install permanent mockup frame fasteners Replace guide pins one location at a time. Install F-04 candidates with correct washers and all-metal locknuts. Keep bolt threads out of primary shear planes where the grip stack permits.	Builder: ____ Inspector: ____
A12	Snug in cross-pattern Bring corner joints to uniform snug condition in a diagonal sequence. Measure frame diagonals again before any final mockup torque.	Builder: ____ Inspector: ____
A13	Apply mockup torque Apply only the schedule's mockup torque where one is stated. For F-04, do not assign a flight torque; use a documented low assembly preload selected by the assembly lead for fit testing.	Builder: ____ Inspector: ____
A14	Install anti-intrusion rails Fit S-008 rails using F-03 candidates, isolation sleeves and large bearing washers. Verify no hard point can contact an inert module block under nominal assembly tolerance.	Builder: ____ Inspector: ____
A15	Install inner fire/thermal pan Place S-002 without wrinkles or sharp unsupported edges. Keep drains and vent entrances open. Bonding jumpers remain mock interfaces unless a bonding test plan is active.	Builder: ____ Inspector: ____
A16	Fit thermal barrier Install S-003 with full coverage and controlled seams. Avoid compression beyond the supplier's provisional limit. Photograph seams before they are hidden.	Builder: ____ Inspector: ____
A17	Install dummy cold plate Fit T-001 with F-07 hardware. Use thickness-controlled dummy interface pads. Verify flatness and that coolant ports face the service boundary.	Builder: ____ Inspector: ____
A18	Install inert module blocks Place numbered inert blocks in the planned segmentation pattern. Install F-06 hold-downs hand-snug, then use the provisional mockup torque. Record each block mass and position.	Builder: ____ Inspector: ____
A19	Install dummy HV/LV components Fit E-002 through E-010 dead envelopes and nonconductive busbars. Preserve service clearances, connector keying and emergency-disconnect access.	Builder: ____ Inspector: ____
A20	Install mock harness and sensors Route W-001/W-002 mock harnesses and I-001 through I-003 sensor bodies. Maintain separation, bend radius, clamp spacing and chafe protection. Tag both ends.	Builder: ____ Inspector: ____
A21	Install dry coolant interfaces Fit T-002/T-003 dry-break envelope models and dry hoses. Verify supply/return keying and ensure no hose crosses the removal path or vent outlet.	Builder: ____ Inspector: ____
A22	Intermediate inspection Inspect concealed fasteners, module restraints, harness routing, thermal barrier coverage, drainage and connector access. Record photographs and discrepancies.	Builder: ____ Inspector: ____

HOLD POINT HP-2: Independent inspection required before closing the sled or installing shield layers.

11. Assembly manual - shield stack, mounts and installation

A23	Fit perimeter frame and seal Install S-011 seal sample without stretching. Verify corners, joint gap and drainage discontinuities match the mockup drawing.	Builder: ____ Inspector: ____
A24	Install crush core Fit S-004 within the perimeter frame. Do not crush during installation. Measure thickness at the defined grid and record any local low spots.	Builder: ____ Inspector: ____
A25	Install anti-penetration layer Place S-005 over the crush core using designated edge supports. If tiled, stagger seams and record tile map/serials. Keep the guarded vent corridor open.	Builder: ____ Inspector: ____
A26	Install standoff/debris frame Fit S-006 using F-02 candidates. Snug in a cross-pattern, verify standoff gap, then apply 20-24 N.m dry mockup torque only after measuring running torque.	Builder: ____ Inspector: ____
A27	Install strike skin Place S-007 using lifting aids if required. Start all F-01 bolts by hand. Snug center-out in a cross-pattern, verify flushness, then apply 7-9 N.m dry mockup torque.	Builder: ____ Inspector: ____
A28	Install impact sensors Mount S-013/I-001 sensor bodies and protected leads. Verify connectors cannot be crushed during shield removal.	Builder: ____ Inspector: ____
A29	Verify vent and drainage Use low-pressure air and smoke-safe visualization only. Confirm flow exits down/aft and no blocked drain or direct line into occupant/gear/duct zones exists.	Builder: ____ Inspector: ____
A30	Install shear-mount cartridges Fit S-009 mock cartridges to the fixture. Use temporary alignment pins, then install F-05 candidates without assigning flight torque. Apply only the test-plan assembly condition.	Builder: ____ Inspector: ____
A31	Install multi-stage latches Fit S-010 latches and secondary retention. Adjust evenly so the sled seats without frame distortion. Confirm visible locked/unlocked state.	Builder: ____ Inspector: ____
A32	Position sled on dolly Center the sled and secure it to the dolly handling fixture. Confirm CG is within the dolly support polygon before movement.	Builder: ____ Inspector: ____
A33	Raise sled into bay Use the lift table in small increments with one lead directing. Stop at any interference. Use guide pins; never use fasteners or latches to pull a misaligned sled into place.	Builder: ____ Inspector: ____
A34	Engage mounts and latches Seat all mount interfaces, remove guide pins, close primary latches, then engage secondary retention. Confirm no single latch carries the assembly.	Builder: ____ Inspector: ____
A35	Connect dummy interfaces Mate blind-mate connector envelopes and dry coolant QDs by hand. Verify full engagement indicators and no forced alignment.	Builder: ____ Inspector: ____
A36	Final torque/locking inspection Audit installed fasteners against F-01 through F-12, record tool IDs and values, apply witness marks where specified, and complete FOD inventory.	Builder: ____ Inspector: ____

A37	Final dimensional survey Record installed envelope, ground clearance, gear/duct/control clearances, sled-to-bay gaps and shield alignment.	Builder: ____ Inspector: ____
A38	Mass and CG update Weigh the complete inert sled and shield, calculate its CG, update vehicle mass properties and compare with the 120 kg system allowance.	Builder: ____ Inspector: ____
A39	Release for inert tests Assembly lead and independent inspector sign the traveler. Release only for the approved inert fit, handling, modal, airflow or low-pressure tests.	Builder: ____ Inspector: ____

HOLD POINT HP-3: The completed inert assembly may not be energized. Any live-cell or HV transition requires a new released configuration, hazard analysis, remote test plan and professional approval.

12. Assembly manual - removal, reinstallation and inspection

R01	Establish safe state For the inert prototype, verify all mock services are disconnected. For any future energized article, use a separate lockout/tagout and zero-energy verification procedure.	Tech: ____ Inspector: ____
R02	Position and chock dolly Center the lift table below the sled, apply wheel brakes and install lateral guides.	Tech: ____ Inspector: ____
R03	Support sled Raise the dolly until supports contact designated pads. Confirm stable four-point support before touching latches.	Tech: ____ Inspector: ____
R04	Disconnect dummy interfaces Unmate dry coolant and connector envelopes by the intended release features. Cap/protect interfaces.	Tech: ____ Inspector: ____
R05	Release secondary retention Remove safety clips/pins in the defined order and account for each item.	Tech: ____ Inspector: ____
R06	Open primary latches Open latches evenly. If binding occurs, stop and correct support height; do not pry or hammer.	Tech: ____ Inspector: ____
R07	Disengage mount interfaces Install guide pins as required, release mock shear-mount retention and verify the sled is fully supported.	Tech: ____ Inspector: ____
R08	Lower in controlled increments Lower while checking all sides. Stop for interference, harness snagging or unexpected tilt.	Tech: ____ Inspector: ____
R09	Roll to service area Lower to transport height, verify CG and restraints, release brakes only when the path is clear, then move at walking speed.	Tech: ____ Inspector: ____
R10	Remove strike skin Support S-007. Remove F-01 bolts in an outside-to-center cross-pattern. Bag/tag hardware by location.	Tech: ____ Inspector: ____
R11	Inspect shield stack Inspect strike skin dents, standoff deformation, core crushing, aramid/ceramic damage, heat/fire-barrier condition, vent blockage and sensor witness data.	Tech: ____ Inspector: ____
R12	Inspect sled Inspect mounts, latches, fasteners, frame, inert module restraints, connectors, harnesses, cold plate and drainage. Record damage with scale and location.	Tech: ____ Inspector: ____
R13	Disposition discrepancies Use accept-as-is, repair, replace or engineering review dispositions. Do not reinstall damaged critical parts without written approval.	Tech: ____ Inspector: ____
R14	Reassemble and reinstall Repeat A23-A39. Replace prevailing-torque nuts when required by the selected hardware specification or when running torque is out of limit.	Tech: ____ Inspector: ____

13. Assembly traveler and sign-off checklist

Record	Entry
Assembly serial / configuration	_____
Drawing/BOM revision	_____
Work order / test article	_____
Assembly lead / technician	_____
Independent inspector	_____
Torque wrench IDs and calibration dates	_____
F-01/F-02/F-03/F-04/F-05 lots	_____
Shield layer serials / material lots	_____
Measured empty sled mass and CG	_____
Measured complete sled mass and CG	_____
Installed clearances	Ground ____ Gear ____ Duct ____ Cockpit ____
HP-1 accepted by/date	_____
HP-2 accepted by/date	_____
HP-3 accepted by/date	_____
Open discrepancies / NCRs	_____
Released inert test scope	_____

Final checks	Pass / Fail / N/A	Inspector initials
No live cells, HV, cryogenics, pressure or pyrotechnics installed	_____	_____
Fastener IDs, grip, locking and mockup torque verified	_____	_____
All concealed-work photographs attached	_____	_____
FOD/tool inventory complete	_____	_____
Vent and drainage paths open	_____	_____
Mounts and latches fully engaged with secondary retention	_____	_____
Ground/gear/duct/cockpit clearances recorded	_____	_____
Dolly and handling fixtures removed or secured	_____	_____
Mass properties updated	_____	_____
Test limitations placarded	_____	_____

14. Inspection plan and acceptance gates

Gate	Method	Preliminary pass condition
Dimensional inspection	CMM/laser tracker or surveyed fixture	Envelope, datums and service clearances match released mockup drawing; deviations recorded.
Mass properties	Calibrated scales and balance fixture	Measured mass and CG entered into aircraft budget; no hidden growth.
Fastener installation	100% traveler review	Correct ID, lot, grip, washer, locking method, torque-tool ID and inspector signoff.
Latch integrity	Repeated cycling and proof pull	Positive lock indication after service cycles; no binding or permanent deformation.
Shield impact coupons	Drop/impactor coupons	No unacceptable intrusion into protected-volume witness block; criteria set before test.
Crush-core response	Quasi-static crush coupons	Stable plateau and energy absorption within selected tolerance.
Vent path	Cold-flow/smoke visualization, then remote fire facility	No cockpit/gear/duct impingement; pressure and temperature remain within derived limits.
Water/dust ingress	Spray/dust test on empty enclosure	No pooling at cells/connectors; drains work; seals remain inspectable.
Modal/vibration	Instrumented shaker on inert sled	No latch release, fastener back-out, fretting, wire chafe or harmful resonance.
Electrical safety	Low-voltage continuity/insulation mock tests first	Interlocks and emergency shutdown logic fail safe; energized testing requires separate hazard review.
Thermal system	Flow bench with heaters and inert modules	Flow balance, leak rate, pressure drop and heat rejection meet requirements derived from component tests.
Containment	Remote destructive test with representative cells	No hazardous propagation or discharge into protected zones; criteria approved before test.

Stop-work triggers: unexpected heating, odor/gas, electrical continuity in an intended isolated zone, cracking/delamination, fastener rotation, latch migration, coolant leakage, unstable dolly motion, or any test result outside the pre-approved envelope.

15. R&D; company work products still required

Discipline	Design/controls/reliability/development output
Requirements	System requirements, interface-control document, hazard analysis, compliance basis, verification cross-reference matrix.
Aerostructures	3D CAD, drawing tree, material allowables, laminate schedules, loads model, FEA, fatigue/damage-tolerance, modal/flutter integration.
Crashworthiness	Crash pulse definition, mount load paths, intrusion limits, occupant-cell interaction, dynamic tests and post-crash egress assessment.
Battery	Chemistry and supplier qualification, cell/module characterization, energy/power sizing, abuse/propagation testing, BMS safety requirements.
Electrical	Voltage architecture, short-circuit/fault-current model, HVIL, creepage/clearance, insulation coordination, grounding/bonding, EMC/HIRF/lightning.
Thermal/fire	Heat-load model, cooling sizing, vent gas characterization, pressure analysis, fire detection/suppression, drainage and materials flammability.
Fasteners/joints	Joint-by-joint load cases, grip stack, preload scatter, torque-tension tests, composite inserts, galvanic isolation, locking and inspection plan.
Maintainability	Removal tooling, lift plan, lockout/tagout, service intervals, connector cycles, damage limits and illustrated parts catalog.
Test	Coupon, subcomponent, sled, iron-bird, environmental, vibration, impact, abuse and containment procedures with calibrated instrumentation.
Quality/configuration	Part numbering, serialized critical parts, approved suppliers, material certs, nonconformance control, drawing revision and software/data configuration.
Certification/safety review	Independent DER/ODA or appropriate certification specialists; FAA engagement strategy; operational limitations; emergency response plan.

Minimum team for the next design phase

Aerospace structures engineer; battery systems engineer; high-voltage power engineer; thermal/fire specialist; crashworthiness engineer; mechanical designer; test engineer; safety/reliability engineer; manufacturing/quality engineer; and certification counsel. No single conceptual drawing can replace these analyses.

16. THE ARC integrated command architecture

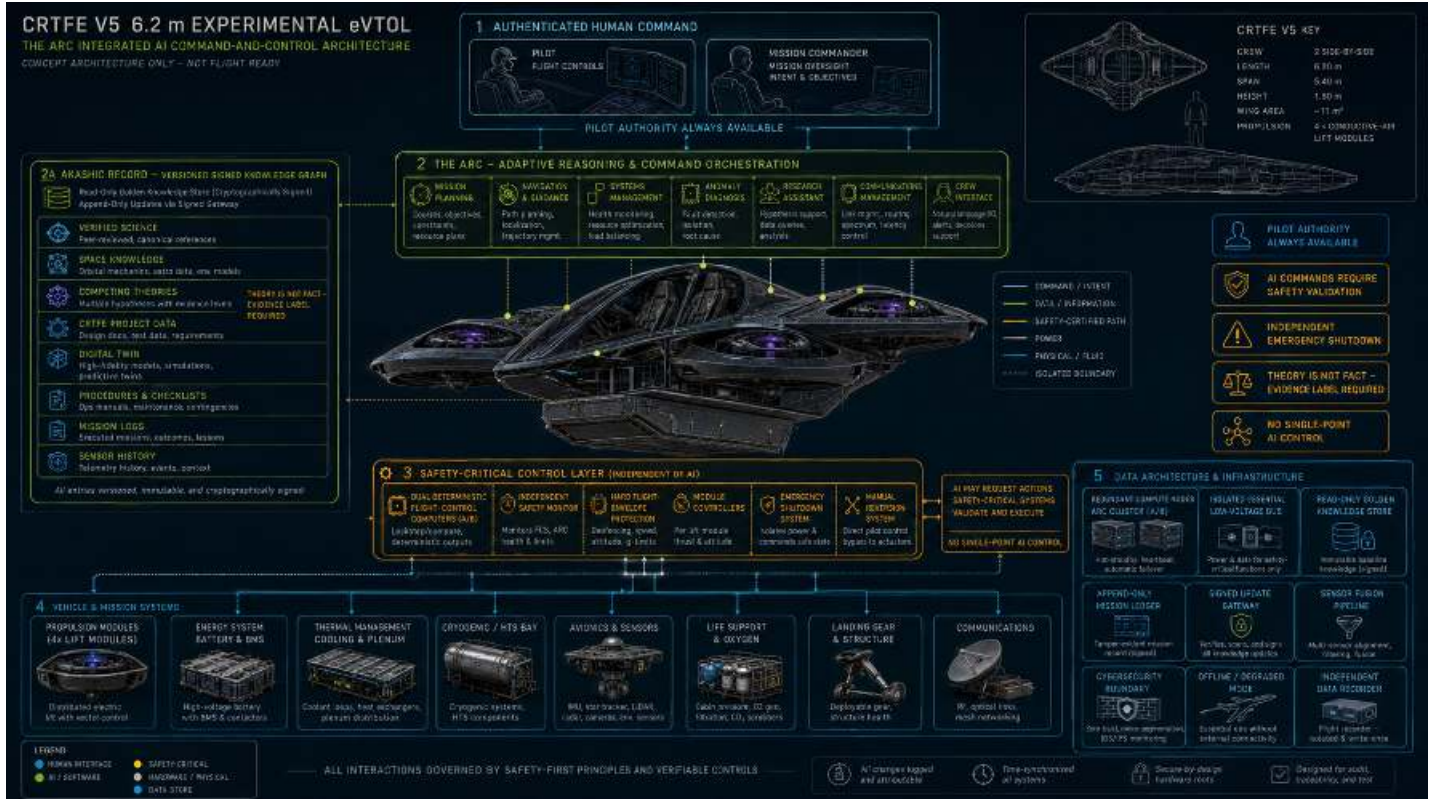


Figure 2. THE ARC vessel-wide command, knowledge and safety architecture. Concept only; not flight-ready software or avionics.

Purpose and authority

THE ARC is the vessel-wide adaptive reasoning and command-orchestration layer. Under authenticated pilot or mission-commander intent, it may plan, recommend, coordinate, diagnose and request actions across the vehicle. It is deliberately prevented from directly driving safety-critical actuators. Dual deterministic flight-control computers, module controllers, an independent safety monitor, hard flight-envelope protection and emergency shutdown validate and execute commands. Pilot authority and manual reversion remain available.

Purpose	Authority source	Permitted role
1	Independent emergency shutdown	Remove propulsion energy and isolate hazardous systems when protected limits are exceeded.
2	Pilot/manual reversion	Direct validated control path independent of THE ARC.
3	Deterministic safety/flight-control layer	Execute control laws, enforce envelopes and reject unsafe or malformed requests.
4	THE ARC	Interpret human intent, optimize mission/system plans and issue bounded requests with rationale and confidence.
5	Akashic Record	Provide signed knowledge, procedures, models, evidence labels and mission context; never command hardware.

No single-point AI control: loss, corruption or disagreement of THE ARC must leave the deterministic control layer capable of continued safe operation or a predefined safe state.

17. Akashic Record knowledge system and ARC bill of materials

Akashic Record definition

The Akashic Record is a technical, cryptographically signed and versioned knowledge architecture - not a claim of supernatural access or omniscience. It stores what humanity and the project currently know, what remains disputed, and why. Every item carries provenance, date, author/source, evidence class, confidence, dependencies, contradictions, applicable vehicle configuration and validation status.

Verified science	Peer-reviewed and canonical references; equations, units, applicability limits, replications and superseded versions.
Space knowledge	Orbital mechanics, astrodynamics, environments, navigation, astronomy, planetary science, communications and human factors.
Competing theories	Hypotheses stored side by side with evidence-for, evidence-against, falsification tests and confidence. Theory never promoted to fact without review.
CRTFE project	Requirements, simulations, measured tests, CAD/digital twin, BOMs, drawings, hazards, procedures and configuration history.
Mission memory	Append-only commands, sensor history, decisions, anomalies, outcomes, crew notes and lessons learned.
Procedures	Normal, abnormal and emergency checklists with approvals, effective configuration and expiration/review dates.
Model registry	AI/model versions, training/evaluation scope, known limitations, safety cases, allowed tools and rollback images.

Preliminary ARC hardware/software BOM

ARC-H01	2	Rugged redundant compute nodes A/B	Dissimilar failure analysis; hot-standby/lockstep strategy; ECC memory; secure boot; independent power feeds.
ARC-H02	1	Independent safety-monitor computer	Separate hardware/software lineage; monitors control requests, health, timing and envelopes.
ARC-H03	2	Read-only golden knowledge stores	Cryptographically signed baseline; physically or logically write-protected in flight.
ARC-H04	2+	Append-only mission ledger storage	Tamper-evident, time-synchronized, fault-tolerant and removable for investigation.
ARC-H05	1	Signed update gateway	Ground-authorized ingestion, malware scan, provenance validation, staging and rollback.
ARC-H06	2	Time/network gateways	Redundant deterministic data buses plus isolated research/crew network.
ARC-H07	1	Independent data recorder	Write-once or protected partition; separate power and crash-survivability study.
ARC-H08	2	Crew interface displays/controls	Clear mode/authority indication, confidence/provenance view, explicit confirmation for consequential commands.
ARC-H09	1 set	Cybersecurity boundary devices	Segmentation, allowlists, rate limiting, intrusion detection and removable-media control.
ARC-S01	1	Command-orchestration software	Typed/bounded command API; rationale; timeout; rollback; no raw actuator access.
ARC-S02	1	Knowledge graph + evidence engine	Provenance, evidence class, contradictions, units, configuration applicability and signed revisions.
ARC-S03	1	Digital twin / state estimator	Real-time vehicle configuration, mass properties, health and predicted margins.
ARC-S04	1	Mission planner and navigator	Constraint-based plans independently checked by certified navigation/control functions.
ARC-S05	1	Anomaly diagnosis engine	Ranked hypotheses, evidence trace, uncertainty, safe troubleshooting actions and escalation.
ARC-S06	1	Offline degraded-mode image	Essential local knowledge/procedures and deterministic interfaces without external connectivity.

18. THE ARC development and verification gates

Gate	Required demonstration before advancement
ARC-0 Knowledge prototype	Ingest project sources; preserve citations; label fact/theory/assumption; retrieve correct configuration-specific procedures; reject unsupported claims.
ARC-1 Desktop digital twin	Operate on simulated vehicle telemetry; produce bounded plans and explain every recommendation with evidence and uncertainty.
ARC-2 Software-in-the-loop	Communicate only through typed command requests; safety monitor rejects out-of-envelope, stale, malformed, contradictory and unauthorized requests.
ARC-3 Hardware-in-the-loop	Run redundant compute, buses, displays and data recorder against an iron-bird. Demonstrate timing, failover, rollback, degraded mode and loss of connectivity.
ARC-4 Fault injection	Corrupt sensors, models, knowledge entries, timing, networks and one compute node. Show safe detection, isolation and continued deterministic control.
ARC-5 Human factors	Pilots can understand authority, uncertainty and rationale; override is immediate; alerts do not overload or mislead; mode confusion is controlled.
ARC-6 Cybersecurity	Independent threat model, penetration testing, signed update validation, key management, supply-chain review and recovery from compromised noncritical nodes.
ARC-7 Ground iron-bird	Coordinate energy, thermal, propulsion and avionics mock systems without direct actuator authority. Record every request and outcome.
ARC-8 Uncrewed bounded test	Only after propulsion/airframe gates, independent safety review and a flight-test authorization. Start with narrow functions and hard geofenced limits.
ARC-9 Crewed consideration	Requires a formal safety case, certification strategy, demonstrated manual reversion and extensive operational evidence. Not authorized by this package.

THE ARC is a research and systems-management concept. It must not be described as knowing everything, predicting unknown physics with certainty, or replacing certified flight controls, engineering judgment or crew authority.

19. Source basis and standards-to-review

Project source. CRFTE V5 Target Vehicle - Concept Definition Package (3 pages). Supplies the 650 kg target, geometry, 120 kg battery allocation, 1.60 m central power bay, 228 kW placeholder load, cryogenic/thermal segregation, and development gates.

Project source. CRTFE V4-V5 Simulation Summary (4 pages). Establishes the conditional nature of the propulsion model, older 450 kg/1.104 kN-per-module baseline, 60 kW-per-module cap, and unresolved conductivity/thermal questions.

FAA AC 20-184. Guidance on testing and installation of rechargeable lithium batteries and battery systems on aircraft. https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentid/1027106

FAA lithium battery resources. FAA material identifies thermal runaway, internal short circuit, overcharge, over-discharge, heat and pressure hazards. https://www.faa.gov/aircraft/air_cert/design_approvals/dah/lithium_batteries

eCFR 14 CFR Part 25. Emergency landing and retention-of-mass concepts are useful design references; the actual certification basis for this vehicle is not established. <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-25>

NASA NTRS. Preloaded joint analysis methodology for space-flight systems provides bolted-joint analysis background. <https://ntrs.nasa.gov/citations/19960012183>

NASA NTRS. Aerospace threaded fastener strength in combined shear and tension highlights the need to analyze combined loading. <https://ntrs.nasa.gov/api/citations/20120003667/downloads/20120003667.pdf>

Standards candidates to procure/review with qualified engineers: RTCA DO-311A (rechargeable lithium battery systems); applicable SAE/NAS/MS fastener specifications; environmental qualification appropriate to the system; wiring, bonding, flammability and structural standards selected through the formal certification basis. Listing a standard here is not a claim of compliance.

Release statement

This P0 package is suitable for design discussion, budgeting, supplier RFIs, CAD coordination and construction of an inert packaging mockup. It is not suitable for procurement of live battery cells, release of flight hardware, structural proof, pressure/fire certification, energized system integration, or any crewed or uncrewed flight test.